

Classification of athletic injuries to muscular tissues

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Athletic muscle injuries

Muscle injuries in sports are very common. For instance, they represent about 30% of all injuries in elite soccer (Ekstrand et al., 2011). The term muscle injury is used here to describe a lesion to the whole muscular tissue complex, including muscle fibers, intramuscular connective tissues (such as the endo- and perimysium), the fascial envelope around the muscle (epimysium), as well as the tendinous extensions. Most athletic muscle injuries affect the lower limb. Even in sports such as tennis and those which are throwing disciplines, injuries of the upper extremity are rare. Age also seems to constitute an important factor. Muscle injuries significantly increase from the age of 25 years upwards. An involvement of connective tissue disruptions in such injuries tends to be associated with longer times needed before returning to play (Renoux et al., 2019). Muscle injuries can have severe consequences for athletic performance, returning to play as well as motivation, and can even lead to financial losses in the case of elite athletes.

Muscles frequently involved in muscle injuries:

- Bi-articular muscles (crossing two joints)
- Muscles with a more complex architecture (e.g., adductor longus)
- Muscles that undergo eccentric contraction
- Muscles that contain primarily fast-twitch type 2 muscle fibers.

The first treatment step in an athletic muscle injury is the establishment of a precise diagnosis, which is crucial for a reliable prognosis. Even although muscle injuries are among the most frequent of all sports injuries (see Chapter 16), they are often evaluated incorrectly and treated inadequately. At the heart of questionable treatment recommendations lies uncertainty, not least because current clinical and imaging diagnosis are inconsistent and not yet sufficiently classified. Consequently, it is frequently difficult to achieve proper assessment of muscle injury and clear communication between practitioners, and the resulting miscommunication can affect rehabilitation progress, recurrence and complication rates.

To facilitate diagnostic, therapeutic and scientific communication, a precise definition of the English muscle injury terminology is needed.

Classification of athletic muscle injuries

Defining and categorizing athletic muscle disorders has traditionally been difficult because they present a heterogeneous group of muscle disorders. Muscles exist in many different shapes and sizes, and their functional and anatomical organization is complex (Armfield et al., 2006). It is therefore challenging to develop a generally applicable terminology and classification of muscle injuries.

While a number of different classification systems have been published in the literature, little

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consistency exists. The limitations of previous grading systems (e.g. by O'Donoghue, Ryan, Takebayashi, Peetrans, Stoller) include the lack of a differentiated spectrum of muscle injuries in athletes. Furthermore, prior grading systems do not include terminology for muscle injuries without macroscopic evidence of structural damage, even although they have high clinical relevance in professional athletes (Ekstrand et al., 2012).

To better reflect the differentiated spectrum of muscle injuries observed in athletes, and to foster discourse centering on the topic of athletic muscle injuries, a new classification system is proposed and introduced in Table 17.1.

The core aspects of the classification system consist of distinguishing between indirect muscle disorders/injuries and direct muscle injuries, and also explicitly recognizing that functional muscle disorders “without macroscopic” evidence (e.g. in MRI or ultrasound) are a category of athletic muscle injury.

Functional muscle disorder

Functional muscle disorders can be described as acute indirect muscle disorders without

macroscopic evidence in MRI or ultrasound. They are often associated with an increase of muscle tone in varying dimensions.

Functional muscle disorders are indirect injuries. They cause more than 50% of the absence of players in professional soccer clubs (Ekstrand et al., 2012). They can also lead to functional limitations. For instance, painful increase of muscle tone can constitute a risk factor for structural injury. However, functional muscle disorders do not show evidence of structural damage in imaging methods such as MRI and are therefore not readily diagnosed with standard diagnostic methods.

Functional muscle disorders can be further distinguished according to the clinical origin of the disorder. There are “overexertion-related” and “neuromuscular” muscle disorders. This distinction is important as it influences the chosen treatment pathway.

Fatigue-induced muscle disorder (type 1A) and delayed onset muscle soreness (type 1B)

Muscle fatigue has been shown to predispose to injury (Opar et al., 2012), for instance, through

Table 17.1 Classification of acute muscle disorders and injuries [adapted from Mueller-Wohlfahrt et al., 2013. Reproduced with permission.]					
A) Indirect muscle disorder/injury				B) Direct muscle disorder	
Functional muscle disorder		Structural muscle injury		Contusion	Laceration
Type 1: Overexertion related muscle disorder	Type 2: Neuromuscular muscle disorder	Type 3: Partial muscle tear	Type 4: (Sub)total tear		
Type 1A: Fatigue-induced muscle disorder	Type 2A: Spine-related	Type 3A: Minor partial muscle tear	Subtotal or complete muscle tear		
Type 1B; Delayed-onset muscle soreness (DOMS)	Type 2B: Muscle-related	Type 3B: Moderate partial muscle tear	Tendinous avulsion		

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increased stiffness. As warming up prior to activity and maintaining flexibility has been shown to decrease muscle stiffness, it is recommended to include these systematically into each training regimen.

Delayed onset muscle soreness (DOMS) is different to fatigue-induced muscle injury. While it occurs several hours after unaccustomed deceleration movements, fatigue-induced muscle disorders can also manifest during athletic activity. As another distinguishing feature, DOMS causes acute inflammatory pain, stiff and weak muscles, as well as pain at rest. It usually resolves spontaneously within a week. By contrast, fatigue-induced muscle disorders lead to aching muscle firmness that increases with continued activity. If it is unrecognized and untreated, it can persist for a longer time, and consequently may cause structural injuries such as partial tears.

Spine-related (type 2A) and muscle-related (type 2B) neuromuscular muscle disorders

Two different types of neuromuscular disorders can be differentiated, (1) a spinal or spinal nerve-related and (2) a neuromuscular endplate-related type.

In the spine-related neuromuscular muscle disorder, functional or structural spinal or lumbopelvic disorders may lead to an irritation of a spinal nerve root and consequently cause an increase of muscle tone.

Back injuries, particularly at the L4/5 and L5/S1 level, are common among elite athletes. It is also well established that pathologies relating to the lumbar spine, such as disc prolapses at L5/S1 level, may present with hamstring and/or calf pain as well as limitations in flexibility. This, in turn, may result in or mimic a muscle injury (Orchard et al., 2004). Such “lumbar spine-related” hamstring injuries are supported by

various studies, although this is a controversial paradigm to researchers.

Against the background sketched above, in the case where a hamstring injury was related to the lumbar spine region, mere treatment of the muscle-tendon unit would not be sufficient. Thus it is important that an assessment of hamstring injury should include a thorough biomechanical evaluation, especially of the lumbar spine, pelvis and sacrum (Woods et al., 2004).

A diagnosis of spine-related neuromuscular disorder is established through clinical functional examination. MRI imaging is usually negative or only shows muscle edema.

The second type of neuromuscular disorder is related to dysfunctions of neuromuscular control mechanisms such as reciprocal inhibition (Figure 17.1).

The neurologic reflex called reciprocal inhibition causes an antagonist muscle to relax when the agonist of a muscle action is contracting. On a neurological level, when sensory information from the agonist muscle is transmitted to the brain, some signals enter the spinal cord, exciting the motoneurons of the associated muscle. Simultaneously, motoneurons of the antagonistic muscles are stimulated via interneurons in the spinal cord, which act via inhibitory synapses.

When the inhibition of the antagonistic muscle is disturbed and agonistic muscles over-contrast to compensate for this, normal muscle tone can be significantly impaired, resulting in painful muscle firmness. This may prevent the athlete from participating in sporting activities.

Structural muscle injury

A structural muscle injury is any acute indirect muscle injury “with macroscopic” evidence of muscle damage in MRI or ultrasound.

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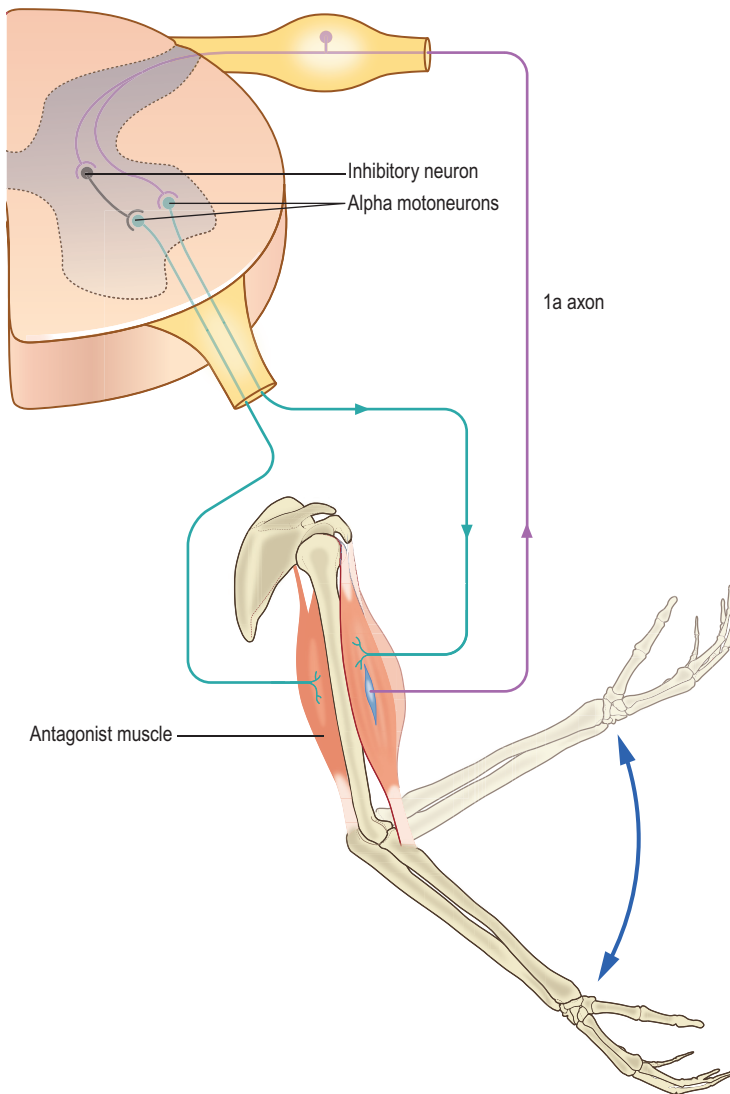


Figure 17.1 Reciprocal inhibition

When sensory information from the agonist muscle (here: M. biceps brachii) is carried to the brain, afferent signals enter the spinal cord on the alpha motoneuron of the agonist muscle. At the same time, interneurons in the spinal cord act via inhibitory synapses on the alpha motoneurons of antagonistic muscles (here: M. triceps brachii).

The most relevant structural athletic muscle injuries are indirect injuries, that is, stretch-induced injuries caused by a sudden forced lengthening over the viscoelastic limits of muscles occurring during a powerful contraction (internal force) (Figure 17.2).

While a tear can theoretically occur anywhere along the muscle-tendon-bone chain, these injuries are usually located at the muscle-tendon junction because these areas present biomechanically weak points. Due to their large

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Figure 17.2

In football/soccer, most muscle injuries affect the four major muscle groups of the lower limbs: hamstrings, adductors, quadriceps and calf muscles; 96% of all muscle injuries in football/soccer occur in non-contact situations (Ekstrand et al., 2011).

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tendinous structures, the quadriceps muscle and the hamstrings are often affected.

Partial muscle tears (type 3)

The majority of indirect structural injuries are partial muscle tears. We distinguish between minor and moderate partial tears. To differentiate between the two can have consequences for therapy, and consequent return to sports. Minor partial tears present a maximum diameter of less and moderate partial tears a diameter of greater than a muscle fascicle/bundle (Figure 17.3).

In addition to the size, it is the participation of the connective tissue that can distinguish a minor from a moderate partial muscle tear. The fascial structures of endomysium, the perimysium, and the epimysium can be affected. Simultaneous injury of the external perimysium seems to play a special role: the external perimysium has a barrier function within the muscle in case of bleeding. It may be the injury to this connective tissue structure that discerns a moderate from a minor partial muscle tear. While minor partial muscle tears may be accompanied with intramuscular edema, intermuscular edema

are more frequently found in moderate partial muscle tears.

Differentiating between partial muscle tears seems difficult because muscles can be structured very differently. Moreover, today's technical capabilities (MRI and ultrasound) are not accurate enough to assess such differences. MRI can be oversensitive and may lead to an overestimation of the actual damage. Further research is needed to be able to differentiate minor from moderate partial muscle tears in the future.

Most muscle injuries heal without forming scar tissue. Greater muscle tears can result in a defective healing with scar formation. Minor partial tears usually heal completely while moderate partial tears can result in a fibrous scar.

(Sub)total muscle tears and tendinous avulsions (type 4)

While complete muscle tears are very rare, subtotal muscle tears and tendinous avulsions are more frequent. Subtotal tears involving more than 50% of the muscle diameter usually heal within the same timeframe as complete tears.

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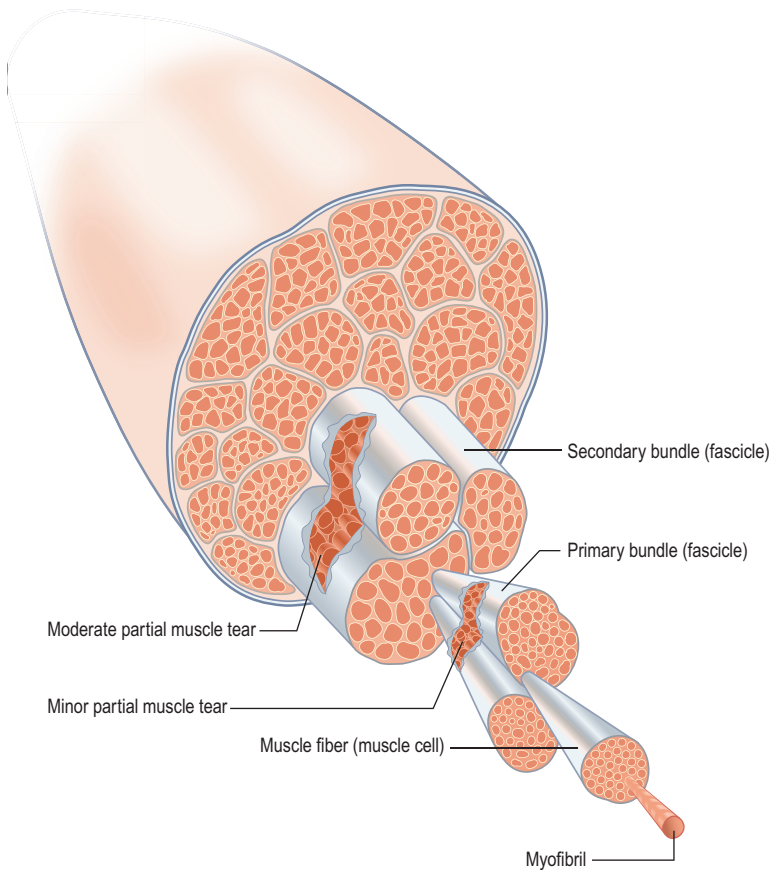


Figure 17.3 Illustration of minor and moderate partial muscle tears in relation to the anatomical structures

Understanding the macro- and microanatomy of musculature is important to correctly classify indirect structural injuries. An individual muscle fiber has an average diameter of 60 μm . Usually, an isolated tear of an individual muscle fiber remains without clinical relevance. Secondary muscle bundles are 2 to 5 mm in diameter, are visible to the human eye, and can be palpated by experienced practitioners when they are torn.

(Adapted from Mueller-Wohlfahrt et al., 2013, with permission).

Biomechanically, tendinous avulsions equal a total tear of the origin or insertion of the muscle. The most often involved sites are the proximal parts of rectus femoris, hamstrings and adductor longus, as well as the distal semitendinosus.

Muscle contusions

Indirect injuries, lacerations or contusions are caused by external forces; these are frequently observed in athletes and present a complex injury that includes defined blunt trauma of the muscular tissue and hematoma (Beiner and Jokl, 2001; Kary, 2010). In clinical practice, it is often

observed that such contusions do not go along with any recognizable tearing injuries to muscular tissues.

Factors affecting the severity of the injury include the contact force and the contraction state of the affected muscle at the moment of injury. Contusion injuries can be classified as mild, moderate or severe. The muscles that are most frequently affected are the rectus femoris and the intermediate vastus. As they lie next to the bone, they have limited space for movement when exposed to a direct blunt blow and are thus prone to contusion injuries. Interestingly, the

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intermediate vastus has been observed to have a very strong tendency towards a degenerative ossification if treated improperly (e.g. with premature loading during a hematoma phase).

Direct injuries such as contusions do not necessarily present with structural damage of muscle tissue. Without such structural damage, athletes may be able to continue competing for a long time, even with more severe contusions. By contrast, an athlete may need to stop immediately when they suffer a fairly small indirect structural injury. However, contusion injuries should not be underestimated. Acute compartment syndrome, active bleeding or large hematomas can be serious and severe complications of contusions.

Diagnosis of muscle injuries

In order to diagnose an athletic muscle injury, we recommend including a precise history of the occurrence. The circumstances and symptoms of the injury as well as previous problems should also be noted. The clinical examination should include inspection, palpation of the injured area, comparison with the unaffected side and functional muscle testing.

Palpation requires experience, sensitivity and intuitive aptitude. The skillful clinical examiner can gather substantial information from palpating the affected area. More superficial and larger tears, particularly of secondary muscle fiber bundles (diameter ~5 mm), as well as perimuscular edema and increased muscle tone, can be detected with seasoned hands.

Practical tip

The physiologic muscle tone varies greatly in athletes. It differs greatly between individuals and depends on training load, type of sport and other factors. It can also vary from side to side.

An ultrasound taken early after the muscle trauma (between 2 and 48 h after injury; Peetrons, 2002) can provide helpful information about any disturbances of the muscle structure. An ultrasound examination is particularly recommended if the clinical examination points towards a functional disorder without the evidence of structural evidence, or if there is any hematoma.

For every injury which is suspicious for structural muscle injury, an MRI examination is recommended. This helps to determine whether and in what pattern edema is present. Furthermore, an MRI can show structural lesions, including approximate size and site of injury. A possible tendon involvement can also be detected via MRI imaging (Askling et al., 2008). However, MRI alone is not sufficiently sensitive to measure the extent of muscle tissue damage accurately. For instance, it cannot be obtained from MRI imaging where edema/hemorrhage is obscuring muscle tissue that has not been structurally damaged.

Practical tip

Changing surfaces, for example, from turf to hall floor or hard courts, leads to an altered demand profile for the musculoskeletal system through, for instance, a receptor-mediated change of muscle tone. The resulting increase in proprioception by the internal structure of the joints inevitably leads to an increase in muscle tone and premature fatigue. This reaction would impede the recovery process after an athletic muscle injury (Figure 17.4). During regenerative training, frequent changes of training surfaces should be avoided.

Typical examples of unnecessary and unfavorable changes of training surface for soccer players include warming up on artificial turf, as well as sprints on all-weather synthetic tracks instead of natural turf.

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Figure 17.4

Injuries are sometimes underestimated or not recognized. A consequent premature return to full training constitutes one possible cause for recurrent athletic muscle injury. Muscle and other connective tissue heal gradually. The immature connective tissue scar is the weakest point of the injured muscle (while mature scars are stiffer or stronger than healthy muscle tissue). Regaining full strength of the injured tissue takes time. Size and localization of the injury also plays a significant role on the healing process.

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